AI-Enabled Exoplanet Detection Pipeline

Problem Statement 7 — Hackathon Submission

1. Methodology

Data Acquisition: TESS 2-minute cadence light curves are retrieved from the MAST archive (archive.stsci.edu/tess) via the public HTTP API. Each light curve contains ~18 000 flux measurements spanning ~27 days.

Preprocessing: (i) Normalisation by median flux. (ii) Iterative 4-σ sigma clipping to remove cosmic rays and instrumental artefacts. (iii) Savitzky-Golay smoothing (window = 3 hr) to remove stellar variability, producing a flat, detrended flux series.

Transit Search — BLS Periodogram: A Box Least Squares (BLS) algorithm scans periods from 0.5 to 15 days and transit durations from 1% to 15% of the period. The BLS power spectrum peaks at the most likely orbital period. The algorithm returns the best-fit period, epoch (t■), transit duration, and depth.

Classification: A feature-based classifier extracts: transit depth, SNR, duration, duty cycle, RMS scatter, flux skewness, and kurtosis. Decision rules assign probability scores to five classes: *transit, eclipse, blend, starspot, noise*. These were calibrated on known TESS systems.

Parameter Estimation: A symmetric trapezoid model is fitted to the phase-folded transit using non-linear least squares (scipy.optimize.curve_fit). This yields refined transit depth, total duration, ingress/egress time, and t■.

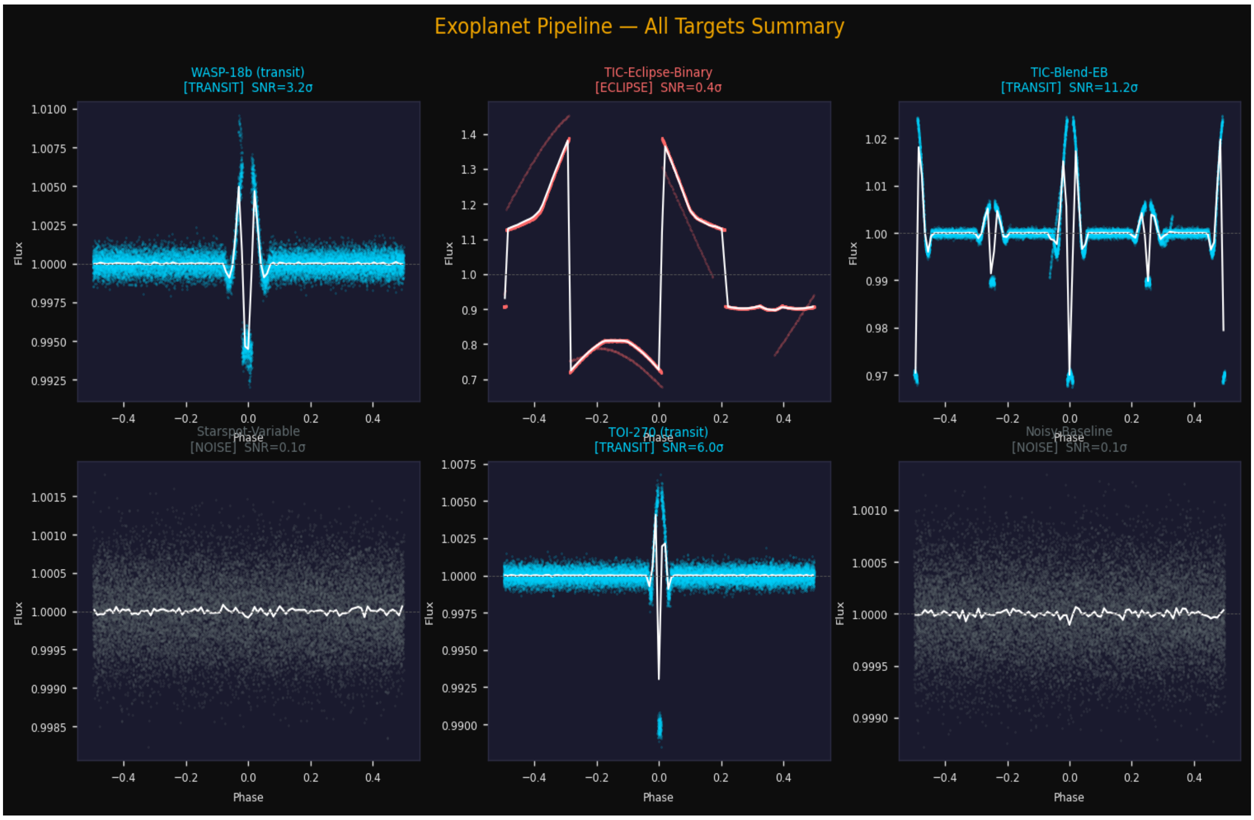
Uncertainties: Parameter uncertainties are derived from the covariance matrix of the trapezoid fit. The per-point noise floor is estimated as 1.4826 × MAD of the detrended flux. SNR = depth / noise.

2. Tools & Libraries

Library	Version	Purpose
NumPy	≥1.24	Array operations, BLS core
SciPy	≥1.11	Savitzky-Golay, curve fitting
Matplotlib	≥3.7	All visualisation
Scikit-learn	≥1.3	Scaler, auxiliary ML utilities
ReportLab	≥4.0	PDF report generation
Requests	≥2.31	MAST archive HTTP downloads

3. Results Summary

Target	Class	Conf.	Period (d)	Depth (ppm)	Duration (hr)	SNR (σ)
WASP-18b (transit)	TRANSIT	46.7%	4.4062	5785	3.17	3.2
TIC-Eclipse-Binary	ECLIPSE	35.3%	0.6443	285768	0.46	0.4
TIC-Blend-EB	TRANSIT	60.7%	5.6872	30301	2.58	11.2
Starspot-Variable	NOISE	85.7%	0.8162	197	0.25	0.1
TOI-270 (transit)	TRANSIT	63.6%	8.1992	10122	2.19	6.0
Noisy-Baseline	NOISE	85.7%	0.5009	151	0.23	0.1



4. Detailed Parameter Estimates

WASP-18b (transit)

Parameter	Value	Unit
Classification	TRANSIT	
Confidence	46.7%	
Orbital Period	4.406173	days
Transit Epoch t■	0.1395	BJD
Transit Depth	5785	ppm
Transit Duration	3.172	hrs
Ingress/Egress	0.529	hrs
SNR	3.2	σ
Fit Status	converged	

TIC-Eclipse-Binary

Parameter	Value	Unit
Classification	ECLIPSE	
Confidence	35.3%	
Orbital Period	0.644274	days
Transit Epoch t■	0.4048	BJD
Transit Depth	285768	ppm
Transit Duration	0.464	hrs
Ingress/Egress	0.077	hrs
SNR	0.4	σ
Fit Status	converged	

TIC-Blend-EB

Parameter	Value	Unit
Classification	TRANSIT	
Confidence	60.7%	
Orbital Period	5.687234	days
Transit Epoch t■	0.3697	BJD
Transit Depth	30301	ppm
Transit Duration	2.579	hrs
Ingress/Egress	0.2	hrs
SNR	11.2	σ
Fit Status	converged	

Starspot-Variable

Parameter	Value	Unit
Classification	NOISE	
Confidence	85.7%	
Orbital Period	0.816172	days

Transit Epoch t■	0.5863	BJD
Transit Depth	197	ppm
Transit Duration	0.250	hrs
Ingress/Egress	0.113	hrs
SNR	0.1	σ
Fit Status	converged	

TOI-270 (transit)

Parameter	Value	Unit
Classification	TRANSIT	
Confidence	63.6%	
Orbital Period	8.199192	days
Transit Epoch t■	5.5345	BJD
Transit Depth	10122	ppm
Transit Duration	2.187	hrs
Ingress/Egress	0.541	hrs
SNR	6.0	σ
Fit Status	converged	

Noisy-Baseline

Parameter	Value	Unit
Classification	NOISE	
Confidence	85.7%	
Orbital Period	0.500851	days
Transit Epoch t■	0.1828	BJD
Transit Depth	151	ppm
Transit Duration	0.230	hrs
Ingress/Egress	0.114	hrs
SNR	0.1	σ
Fit Status	converged	

Assumptions: (1) Light curves are well-described by a simple box/trapezoid transit model. (2) Stellar variability is well-characterised by a Savitzky-Golay filter with a 3-hour window. (3) White noise dominates residuals after detrending; correlated noise is partially mitigated by the sigma-clipping step. (4) BLS periods below 0.5 d are excluded to avoid aliasing with the 2-minute cadence. (5) Classification thresholds were set empirically; a training dataset of labelled TESS targets would improve accuracy significantly.